

Bash Scripting

🌀 1. What is Bash Script?

A Bash script is a file of commands that run automatically.

```
#!/bin/bash  
echo "Hello World"
```

📁 2. Variables

Used to store data.

```
name="Umar"  
age=20
```

```
echo $name  
echo $age
```

- ✓ No space around =
 - ✓ Use \$ to access value
-

⬇️ 3. Arguments (Command Line Input)

```
#!/bin/bash  
echo "Name: $1"  
echo "Age: $2"
```

Run:

```
./script.sh Umar 20
```

🔗 \$1, \$2 = inputs

🔗 \$# = number of arguments

4. User Input

```
Echo -ne "Enter your name:"  
read name  
echo "Hello $name"
```

5. If Condition

```
num=10  
  
if [ $num -gt 5 ]  
then  
    echo "Greater than 5"  
fi
```

6. If-Else

```
num=8  
if [ $num -gt 5 ]  
then  
    echo "Greater"  
else  
    echo "Smaller"  
fi
```

🔄 7. If-Elif-Else

```
num=6
if [ $num -lt 5 ]
then
    echo "Less"

elif [ $num -eq 6 ]
then
    echo "Equal"

else
    echo "Other"
fi
```

🔧 8. Operators

Numbers:

- -eq equal
- -ne not equal
- -gt greater
- -lt less
- -ge, -le

Strings:

```
if [ "$name" = "Umar" ]
then
    echo "name matched"
```

+ 9. Mathematical Operations

```
a=5
b=3

sum=$((a + b))
echo $sum
echo $((10 * 5))
```

10. Functions

```
greet() {
    echo "Hello $1"
}

greet Umar
```

© 11. Case Statement

```
read choice

case $choice in
    1) echo "One";;
    2) echo "Two";;
    *) echo "Invalid";;
esac
```

12. Menu Example

```
echo "1. Show Date"
echo "2. Show User"
read ch

case $ch in
    1) date;;
    2) whoami;;
    *) echo "Wrong option";;
esac
```

13. For Loop

```
for i in 1 2 3
do
    echo $i
done
```

14. While Loop

```
i=1
while [ $i -le 3 ]
do
    echo $i
    ((i++))
done
```

15. Until Loop

```
i=1
until [ $i -gt 3 ]
do
    echo $i
    ((i++))
done
```

16. File & Directory Conditions

```
if [ -f "file.txt" ]
then
    echo "File exists"
fi

if [ -d "folder" ]
then
    echo "Directory exists"
fi
```

17. Pipes & Redirection

```
cat file.txt | grep "error"  
echo "Hello" > file.txt
```

18. Permissions

```
chmod +x script.sh
```

19. Arrays

1. Creating an Array

```
names=("Umar" "Ali" "Ahmed")
```

2. Accessing Array Values

```
echo ${names[0]}    # Umar  
echo ${names[1]}    # Ali
```

 Index starts from 0

3. Print All Values

```
echo ${names[@]}
```

 Output:

```
Umar Ali Ahmed
```

12 34 4. Length of Array

```
echo ${#names[@]}
```

🔗 Output:

3

+ 5. Add Values to Array

```
names+=("Zaid")
```

✕ 6. Remove Value

```
unset names[1]
```

🔗 Removes "Ali"

↻ 20. Loop Through Array

```
names=("Umar" "Ali" "Ahmed")
```

```
for name in "${names[@]}"  
do  
    echo "hello how are you $name"  
done
```

↓ 21. Array with User Input

```
read -a names  
echo ${names[@]}
```

22. Arrays in Functions

```
print_names() {  
    arr=("$@")  
  
    for n in "${arr[@]}"  
    do  
        echo $n  
    done  
}  
  
names=("Umar" "Ali")  
  
print_names "${names[@]}"
```

23. Loop through array using user input

```
echo -ne "Enter first name : "  
read firstname  
echo -ne "Enter second name: "  
read secname  
echo -ne "Enter third name: "  
read thirdname  
names=($firstname $secname $thirdname)  
for j in "${names[@]}"  
do  
    echo "hello mr.$j"  
done
```

⚠️ Important Tips

- ✓ Use spaces inside []
- ✓ Use quotes " " for strings
- ✓ Use "\${array[@]}" for arrays